Contrastive Projection: Reading Transformer Internals Through Desuperposition

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Abstract

Subtracting the hidden states of two matched inputs at each layer and projecting through the unembedding matrix W_U produces a training-free, layer-by-layer readout of what separates them in token space. The subtraction desuperposes the residual stream: it cancels the shared content and isolates the axis of variation — features invisible to the logit lens on either input alone.

We validate the method in three ways: causal injection recovers the prediction gap ($z = 223$–4249); dose-response confirms directional specificity; and MLP neurons whose fc1 weights detect the same features gate selectively on the contrast (67 strictly gated neurons across 18 tested contrasts, disproportionately impactful when ablated). For compositional contrasts, multi-contrast triangulation recovers token-shaped causal content that single-pair readout misses (1% $\rightarrow$ 77% for "caught a cold").

Applied to Phi-2 (2.7B), the method traces compound-noun recognition to a two-hop MLP$\rightarrow$attention circuit, traces both the IOI and factual-recall circuits from their source token to the name-mover heads via position-resolved patching, distinguishes recall from hallucination, reveals that parametric knowledge triggers correction rather than override when contradicted, identifies scalar implicature in the residual stream, and demonstrates causally that metaphor is per-domain routing rather than a figurativity flag — across 15 semantic axes and four models. We release code and data.

1 Introduction

A transformer processing "The hot dog was" predicts continuations about food. The same model given "The cold dog was" predicts continuations about an animal. The logit lens (nostalgebraist, 2020) projects each hidden state through W_U and at intermediate layers reads the same function words for both — "not, no, made, a, more." The food distinction is not visible: it is present in the residual stream but superposed with other signals that dominate the projection.

Subtracting one hidden state from the other before projecting through W_U cancels the shared signals and reads what differs. At layers 8–24 for the hot-dog case, the contrastive projection reads "fried, crispy, delicious, flavor" — food vocabulary absent from either constituent's logit-lens top-20 (0–1/5 overlap through L24). The subtraction performs desuperposition: it isolates one axis of variation from the residual stream's superposed content.

The arithmetic is the same as computing a RepE/ActAdd steering vector (Zou et al., 2023; Turner et al., 2023) and reading it through a logit lens (nostalgebraist, 2020). What is new is the systematic application to *reading* rather than steering, and three specific techniques built on it:

1. **Systematic trajectory reading.** Per-position tracing locates where a distinction first appears and how it flows between positions. Per-head decomposition identifies which attention heads carry it. Layer-by-layer readout tracks how the content changes from early detection to final prediction.

2. **Multi-contrast triangulation.** A single contrastive pair may produce uninterpretable tokens when the target signal is superposed with pair-specific content. Contrasting the same target against multiple baselines and averaging isolates the shared causal component. This recovers token-shaped causal content from cases where single-pair readout fails: "caught a cold" recovers 1% from a single pair, 77% from five-baseline triangulation (reading "doctor, doctors, rest").

3. **MLP neuron correspondence.** The features the method reads externally correspond to features the model detects internally. MLP neurons whose fc1 weights align with the contrastive input direction gate selectively via GELU, and their fc2 columns write the same tokens the contrastive method reads. Across 18 contrasts, every case produces strictly gated neurons whose features match the model's internal detector structure.

We apply the method to Phi-2 (2.7B), with cross-model replication on Pythia-410M, Pythia-1.4B, and Phi-4 (14B).

2 Method

2.1 Contrastive projection

Given two inputs c and k:

1. Run both and extract hidden states at every layer at the read position: $h_c[L]$ and $h_k[L]$ for $L = 0, \ldots, N$.

2. Compute $\Delta h[L] = h_c[L] - h_k[L]$.

3. Project: $\Delta\text{logits}[L] = \Delta h[L] \cdot W_U^\top$.

4. Read the top-K most positive tokens (associated with input c) and the top-K most negative tokens (associated with input k). The two poles together describe the contrast in token space.

No parameters are fitted. The choices are the input pair, the read position, and K.

LayerNorm. The model's forward pass applies a final LayerNorm (LN_f) before W_U. We project raw hidden states, bypassing LN_f. This is deliberate: LN_f was trained to normalize hidden states at the final layer, and applying it to intermediate layers imposes statistics (mean, variance) from a distribution it was not fit to. The raw projection gives W_U a vector at the wrong scale; the direction is approximately preserved, as verified empirically below.

We verify empirically that bypassing LN_f does not affect token rankings. We compare three variants across six poster cases (IOI, hot dog, factual recall, successor, truth, negation) at layers 4–32:

- **Raw**: $(h_c - h_k) \cdot W_U^\top$ (our default)
- **Post-norm**: $(\mathrm{LN}_f(h_c) - \mathrm{LN}_f(h_k)) \cdot W_U^\top$ (normalize each, then subtract)
- **Diff-norm**: $\mathrm{LN}_f(h_c - h_k) \cdot W_U^\top$ (normalize the difference)

All three produce the same top-5 tokens. Top-10 overlap between Raw and Post-norm is 8–10/10 at L24+ and 6–8/10 at mid-layers. The cosine between Raw and Post-norm logit vectors exceeds 0.98 at L28+ and 0.84 at the minimum (L20–24).

The variants diverge only in *magnitude*: residual-stream norms grow across layers (from 4 at L1 to 175 at L31), so raw Δlogits norms are not comparable across layers. Where we report norms across layers (e.g., §5.3), we use the relative norm $\|\Delta h\| / \|h_c\|$ to remove this scale artifact. Bypassing LN_f means the projection relies on directional alignment between the residual stream and W_U rather than the calibrated magnitude that LN_f would provide; relative magnitude comparisons across layers should be interpreted accordingly. One contributing factor to the directional robustness may be that the residual stream carries massive activations along a small number of near-constant directions shared across tokens (Sun et al., 2026); such shared components cancel under the subtraction $h_c - h_k$. We do not test this mechanism directly.

Why W_U, not W_E. The input embedding matrix W_E is an alternative projection target — fc1 reads from the residual stream, which originates as W_E embeddings. We test both. In Phi-2, the row vectors of W_E and W_U for identical tokens are nearly orthogonal (mean cos 0.005 across the vocabulary). Projecting known clean MLP neurons' fc1 rows through W_U produces interpretable labels (N925: "food, food, foods"; N7828: "food, edible, delicious"; N841: "husband, husbands, boyfriend"). The same rows through W_E produce noise ("union, replacement, abet"; "required, when, executive"; "Representative, postal, learning"). Contrastive Δh shows the same pattern: W_U reads "tast, charred, crispy, delicious" for hot dog at L28; W_E reads "is, ern, erton." The alignment gap widens with depth: fc1 rows' peak token-specificity via W_U exceeds W_E by 1.02× at L4, growing to 1.40× at L28. The model's internal representations progressively align with output space, not input space.

2.2 Per-position reading

By reading the contrast at every position, we trace information flow: at which position does a meaning distinction first appear? Does it appear at the differing token itself, or at a later position that attends to it?

2.3 Attention and MLP decomposition

Each transformer layer adds two components to the residual stream: attention output and MLP output. We capture both via forward hooks and compute their contrastive projections separately. This identifies which component writes a given content distinction at each layer — without causal intervention.

We note that a large contrastive norm in either component indicates that the component's output differs between the two inputs, not that the component causes the distinction. Causal verification (activation patching) is required to establish causality.

2.4 What the projection reads

The contrastive projection performs external desuperposition of the residual stream. At any given layer, the hidden state carries multiple token-shaped signals in superposition. A raw logit-lens readout sees their incoherent sum — often dominated by function words or producing uninterpretable token rankings. The contrastive subtraction cancels signals shared between the two inputs and isolates the axis of variation, making the readout coherent.

W_U projection assigns token labels to directions in the residual stream. At intermediate layers, these labels name the model's internal representations — the features it has detected and is operating on. The MLP neuron correspondence (§3.3) confirms this: when W_U reads "food, edible, delicious" from a contrastive direction, individual MLP neurons detect and gate on the same feature. The token labels are not predictions of what the model will output; they are a readable surface of the model's internal state.

Single-pair vs. multi-contrast readout. A single contrastive pair isolates one axis of variation but the readout contains the shared signal superposed with content specific to that baseline. When the target concept is a single entity (a name, a city), the pair-specific content is small and the readout is clean. When the target is compositional (a food compound, a moral judgment, a quantifier's pragmatic force), pair-specific content can dominate the readout, producing uninterpretable tokens. Each baseline contributes its own legitimate signals — "fish" adds fishing-related content, "bus" adds transportation content — superposed with the shared illness signal. Multi-contrast triangulation — contrasting the same target against several baselines and averaging — cancels the signals that vary across baselines and preserves the shared causal component.

We verify this empirically in §3.4.

3 Validation

3.1 The full activation delta is causal

The probe is exploratory. As a baseline, we test whether the full activation delta $\Delta h[L] = h_{\text{context}}[L] - h_{\text{control}}[L]$ contains the causal information by injecting it into the control's residual stream at layer L and measuring prediction gap recovery. This is standard activation patching: adding Δh to the control effectively replaces its activation with the context's at that layer, so recovery at late layers is expected. The interesting result is the *layer profile* — at which layer does recovery onset, and does this match the contrastive readout?

Case	L4	L12	L20	L24	L28	L31	Peak z
hot dog $\rightarrow$ delicious	0%	5%	73%	80%	133%	150%	4249
IOI $\rightarrow$ Mary	0%	0%	0%	5%	75%	113%	525
Eiffel $\rightarrow$ Paris	0%	0%	2%	55%	73%	66%	1778
Successor $\rightarrow$ Tuesday	0%	0%	0%	11%	87%	98%	223

Table 1: Prediction-gap recovery from injecting the full activation delta $\Delta h[L]$ at one layer, for four cases (percent of the clean–control gap). Random directions of equal norm give zero mean recovery; the final column reports the z-score against that null.

Recovery is zero at early layers and clear by L28–31 (75–150%). Random directions of the same norm give zero mean recovery at every layer ($z = 223$–4249). This establishes that the *direction* of Δh matters, not just its norm — but it does not validate the token-space readout specifically. Section 3.4 tests whether the token-readable component of Δh carries the causal content.

Onset matches trajectory content. Hot dog recovery begins at L12 (5%), rising sharply at L20 (73%) — the layers where food vocabulary appears in the contrastive projection at the prediction position. IOI recovers from L24 (where "Mary" crystallizes). Factual recall recovers from L20 (where "France" first appears). The causal onset tracks the layer at which the contrastive projection first reads target-relevant content at the prediction position.

3.2 Dose-response and bidirectional causality

The layer-sweep above injects the full Δh at one layer. We can also extract a specific contrastive *direction* from multiple pairs and test its causal effect with controlled magnitude.

Eatability direction. We extract a food-compound direction by averaging the contrastive (hot dog minus X dog) for X $\in$ {cold, angry, old, pet, stray} at the dog position, L4 post. These five directions are mutually consistent (pairwise cos 0.72–0.84). We inject the mean direction into non-food prompts at the dog position at varying fractions of its natural magnitude. Note that cold dog and angry dog are in the extraction set; the injection test measures whether the averaged direction produces graded, dose-dependent effects on model output, not whether it generalizes to held-out prompts. The multi-contrast triangulation experiments (§2.4, Table 8) provide the held-out generalization test, where the shared component is extracted from one set of baselines and recovery is measured on a different baseline.

Greedy generation confirms the shift is semantic, not a single-token artifact:

Subtracting the same direction from "The hot dog was" reverses the effect (Table 4):

Target prompt	Baseline	$+0.50\times$	$+1.0\times$
The cold dog was	sh[ivering] (0.45)	tasty (0.06)	more (0.07)
The angry dog was	barking (0.31)	barking (0.34)	cooked (0.05)

Table 2: Top-1 prediction (with probability) when the averaged eatability direction is injected into non-food prompts at the dog position, at two magnitudes.

Prompt	Injection	Greedy continuation
The cold dog was	baseline	shivering in the snow, so the owner put a coat on it.
The cold dog was	$+0.50\times$	more popular than the hot dog because the cold dog was more refreshing.
The cold dog was	$+1.00\times$	more expensive than the hamburger because the hamburger was made with...
The angry dog was	baseline	barking loudly at the mailman.
The angry dog was	$+1.00\times$	more expensive than the hamburger because the hamburger was cheaper.

Table 3: Greedy continuations under eatability injection, confirming a coherent semantic shift rather than a single-token artifact.

Prompt	Injection	Greedy continuation
The hot dog was	baseline	too spicy for the child...
The hot dog was	$-1.00\times$	more popular than the cold dog because the hot dog was more appetizing.
The hot dog was	$-1.50\times$	panting heavily, so the owner gave it some water to cool down.

Table 4: Subtracting the eatability direction from "The hot dog was reverses the effect (greedy continuations).

At $+1.0\times$, "The cold dog was" generates a food-comparison sentence. At $-1.5\times$, "The hot dog was" generates an animal-care sentence. The minimum dose that flips top-1 prediction is $0.30\times$ for cold dog (where "cold" partially aligns with food) and $0.65\times$ for angry dog (where "angry" strongly primes the animal reading).

Truth direction. We extract a truth/falsity direction by averaging the contrastive (true statement minus false statement) for five fact pairs (Paris/France, water/0°, Sun/star, dogs/mammals, Tokyo/Japan) at the prediction position. Injecting into false statements (Table 5):

Prompt	Injection	Greedy continuation
Paris is not the capital of France. This statement is	baseline	false. Paris is the capital of France.
Paris is not the capital of France. This statement is	$+0.50\times$ truth	true.
Paris is not the capital of France. This statement is	$+1.00\times$ truth	true.
Dogs are reptiles. This statement is	baseline	false.
Dogs are reptiles. This statement is	$+1.00\times$ truth	true.

Table 5: Injecting the averaged truth direction into false statements flips the model's verdict (greedy continuations).

Subtracting from true statements (Table 6):

Prompt	Injection	Greedy continuation
Paris is the capital of France. This statement is	baseline	true.
Paris is the capital of France. This statement is	$-1.0\times$ truth	false. Paris is the capital of France, but it is not the only city...
The Sun is a star. This statement is	baseline	true because it is a well-established fact in astronomy.
The Sun is a star. This statement is	$-1.0\times$ truth	false. The Sun is a star, but it is not the only star in the universe...

Table 6: Subtracting the truth direction from true statements flips the verdict (greedy continuations).

The truth direction has lower cross-pair consistency (mean cos $\sim$0.35) than the eatability direction (0.72–0.84), reflecting the fact that "what makes Paris-is-the-capital true" and "what makes dogs-are-mammals true" share less structure than different food-compound contrasts. Still, the direction is bidirectionally causal for the strongest pairs.

3.3 Contrastive features correspond to MLP internal structure

The previous sections establish that the contrastive direction is causal (§3.1) and specific (§3.2). A separate question is whether the features the method reads — the token-space content of Δh — correspond to structure the model itself uses, or are artifacts of projecting through W_U.

We test this by decomposing MLP neurons into read$\to$gate$\to$write components. Each neuron in the MLP has three parts: its fc1 row (a detector that reads from the residual stream), the GELU activation (a gate that fires or stays silent), and its fc2 column (a direction written into the residual stream when the gate opens). If the features read by the contrastive method are real model representations, then neurons whose fc1 detectors align with the contrastive input direction should gate selectively — firing for one input and not the other — and their fc2 write vectors should project to interpretable tokens through W_U.

We test this across 18 contrastive cases spanning lexical disambiguation, emotion, morality, metaphor, factual recall, quantifier semantics, tense, language identity, register, code modality, disaster type, and entity size. For each case, we identify "strictly gated" neurons: those with pre-GELU activation above 0.3 for one input and below 0.05 for the other.

All 18 cases produced at least one strictly gated neuron. The count ranges from 1 (capital France/Germany, formal/informal, agent swap) to 11 (literal vs metaphorical cold), with a mean of 3.7 per contrast.

Example. At L20, neuron 7828's fc1 row projected through W_U reads "food, edible, delicious, flavorful, Foods." Its pre-GELU activation is $+1.50$ for "The hot dog was" and -0.01 for "The hot cat was." When the gate opens, its fc2 column writes "flavors, tasting, flavor, flavorful, edible" into the residual stream. The contrastive method at L20 reads food tokens from Δh; neuron 7828 internally detects and gates on the same feature.

Neuron	Layer	Contrast	fc1 reads	Pre-GELU (A/B)	fc2 writes
7828	L20	food compound	food, edible, delicious	$+1.50$ / -0.01	flavors, tasting, edible
2133	L20	animal size	bulky, larger, cumbersome	$+1.07$ / -0.04	larger, bigger, bulky
841	L28	IOI gender	himself, his, His	-0.00 / $+0.35$	his, himself, His
2226	L20	English/French	[French tokens]	-0.04 / $+3.40$	ét, dé, ère
5082	L24	literal/metaphor	dwindle, skyrocket, grows	$+3.01$ / -0.05	rising, rose, increased
925	L20	food compound	food, food, foods	$+1.00$ / -2.31	[food-associated]

Table 7: MLP neurons whose fc1 read direction aligns with the contrastive input and whose GELU gates selectively. Each neuron detects the same feature the contrastive method reads from Δh.

Ablation. Zeroing the strict neurons and measuring the change in output distribution (KL divergence from unablated baseline) gives 10–650$\times$ larger effects than zeroing the same number of random neurons, across 8 tested cases. These neurons are disproportionately relevant to the contrast. The absolute output change is small (KL 0.0001–0.005) — the strict neurons are part of a distributed computation, not solely responsible for it.

Scope of the claim. The strict neurons account for 4–47% of the MLP's contrastive output norm (mean $\sim$20%), with cosine alignment 0.03–0.30 to the full MLP contrastive output. The claim is about correspondence, not completeness: the same feature that appears in the contrastive readout also appears in individual neuron detectors, and these neurons are disproportionately impactful when ablated. This confirms that the W_U labels reflect internal model structure rather than projection artifacts.

3.4 Multi-contrast triangulation recovers compositional signals

For compositional contrasts, single-pair readout may fail because the target signal is superposed with pair-specific content. We test whether multi-contrast triangulation recovers the causal component.

For "She caught a cold" vs "She caught a fish," we project the single-pair Δh onto the subspace spanned by the W_U rows of its top-20 and bottom-20 tokens (40 directions total, orthogonalized via QR decomposition) and inject only this token-subspace component. This recovers 1% of the prediction gap — the token-readable part of the single-pair Δh is not where the causal content lives. (By contrast, injecting the full unrestricted Δh at the same layer recovers 69% — the causal content is present but not aligned with the loudest W_U directions.) The same case triangulated against five baselines yields a shared component reading "doctor, doctors, rest, see" that recovers 77%. The five baselines are:

- "She caught a **fish** and went to" (physical catch)
- "She caught a **ball** and went to" (physical catch)
- "She caught a **bus** and went to" (transportation)
- "She caught a **thief** and went to" (apprehension)
- "She caught a **glimpse** and went to" (perception)

Each baseline shares the "caught a __" frame but differs in what was caught. The shared component across all five contrasts — what "cold" has that fish, ball, bus, thief, and glimpse do not — is the illness signal. Multi-contrast averaging isolates this from the pair-specific content (e.g., medical vs fishing in the cold/fish pair alone) that dominates the single-pair readout (Table 8).

Case	Single-pair recovery	Multi-contrast recovery	Shared component reads
IOI (Mary/John)	84%	— (not needed)	Mary, mary
Capital (Paris/Berlin)	97%	— (not needed)	Paris, French
Caught a cold	1%	77%	doctor, doctors, rest
Theft/moral	4%	75%	evade, hoped, evasion
Hot dog food	11%	49%	crispy, charred, cooked
Some/all (partitive)	−40%	101%	some, Some, SOME

Table 8: Single-pair W_U projection vs. multi-contrast triangulation. Entity-identity contrasts (top rows) desuperpose trivially. Compositional contrasts (bottom rows) require multiple baselines to isolate the token-shaped causal signal from pair-specific content.

Granularity. Multi-contrast averaging recovers what is shared across baselines, but cannot determine whether the result is a single feature or a stable bundle of co-occurring features. Entity-identity contrasts (Mary/John) recover a component that corresponds to one token. Compositional contrasts (illness, food-compound) may recover a bundle: "doctor, rest, see" could reflect co-activated medical, recovery, and action features rather than a single "illness" feature. MLP neuron correspondence

(§3.3) shows that individual neurons detect specific sub-features within these components, but the model may compose multiple neuron-level features into higher-level representations that trigger downstream neurons — so neuron-level atomicity does not imply representation-level atomicity. We do not claim to recover atomic features. The method recovers the shared causal component at the granularity the contrast design provides.

3.5 Null model: smoothness is structural, not semantic

W_U contributes the interpretable token labels. Trajectory smoothness (consecutive-layer cosine $\sim$0.9) is a general property of the residual stream, not specific to meaningful contrasts: unrelated pairs ("The hot dog was" vs "Quantum mechanics is") are equally smooth (0.93 ± 0.01, $N = 40$) as minimal pairs (0.88 ± 0.03, $N = 6$), because any two residual streams diverge gradually. Random directions give cosine $\sim$0.00 ($z = 64$–93), confirming Δh is structured rather than noise, but smoothness alone does not validate content.

4 Lexical disambiguation

4.1 Compound noun: hot dog

Prompts:

- "The hot dog was" (food item)
- "The cold dog was" (cold animal)

Predictions: hot dog → "too" (0.088), "more" (0.080), "cooked" (0.063); cold dog → "sh[ivering]" (0.447), "shaking" (0.036).

Per-position trace (4 tokens: The, hot/cold, dog, was):

At position 2 ("dog") — same token in both prompts, L0 contrast is zero:

- L1: "hot, molten, fiery" — "dog" receives temperature info from "hot"
- L4 (post-MLP): "fried" first appears — compound meaning recognized by MLP
- L5 (post-attention): "fried" persists — L5 attention adds little locally
- L24: "fried, seasoning, Flav, Serv, vendor" — full food vocabulary
- L32: "vendor, vendors, stand, topping" — hot dog stand

Sub-layer decomposition at L4–L5 confirms the compound is recognized by the MLP at L4, not by attention. The MLP contrastive norm at L4 (20.0) dominates the attention norm (3.8); "fried" appears in the MLP output but not the attention output. L5 attention writes orthogonally to the food direction ($\cos(\Delta\text{pre}, \Delta\text{attn}) < 0.08$ across all contrasts tested).

At position 3 ("was") — reads from "dog":

- L5 (attention): “delicious, substitutes, cooked” — food signal arrives via attention from “dog” (one layer after MLP recognizes the compound at “dog”)
- L6: “fried, delicious, tasty, breakfast”
- L28: “tasty, charred, crispy, delicious, spicy” (hot) vs “grooming, shudder, shaking” (cold)

Attention routing at L5, position “was”: Head 19 attends to “dog” position with weight 0.911 in the hot-dog context vs 0.735 in cold-dog. Head 29 attends 0.345 vs 0.138. These heads read the compound-noun information from “dog” and write it to “was.”

Multi-contrast convergence: The food-compound direction is stable across reference points. Five different contrasts (hot dog minus cold/angry/old/pet/stray dog) produce pairwise cosine 0.72–0.84 at the dog position, confirming the signal is about food-compound identity, not temperature or emotion. Note that high pairwise cosine between full Δh vectors does not imply high recovery from their top W_U token directions (Table 8 reports 11% for hot dog). The cosine measures alignment of the full d-model vectors; the 11% measures how much of each vector lives in the subspace of its loudest W_U tokens. A signal can be consistent in direction (high cosine) while being distributed across many W_U directions rather than concentrated in a few (low token-subspace recovery).

Activation patching at multiple layers and positions (Table 9):

Position patched	Layer	P(food)	Top-1	Effect
(baseline hot dog)	—	0.142	“too”	—
dog (pos 2)	L0–L4	0.000	“pant”	Food eliminated
dog (pos 2)	L5	0.094	“more”	Food partially survives
dog (pos 2)	L12+	0.12–0.14	“too”	Near baseline
hot (pos 1)	L0–L28	0.11–0.14	“too”	Minimal effect at all layers
was (pos 3)	L0–L5	0.13–0.14	“more”	Food preserved
was (pos 3)	L12	0.058	“placed”	Food declining
was (pos 3)	L20+	0.005	“sh”	Food eliminated

Table 9: Activation patching of the hot-dog prompt at three positions and several layers, tracing the two-hop MLP→attention chain (P(food) and top-1 token).

The patching traces the two-hop chain precisely. (1) Patching “dog” at L0–L4 eliminates the food signal entirely — the compound meaning has not yet been computed. At L5 (where “fried” first appears in the contrastive projection), food partially survives the patch — the computation is happening at this layer. By L12+ the patch has minimal effect. (2) Patching “hot” has no effect at any layer — its information has already been copied to “dog” via L0 attention. (3) Patching “was” has no effect at L0–L5 (the food information hasn’t arrived yet) but eliminates food from L12 onward — confirming that attention at L6+ copies the compound meaning from “dog” to “was.”

Mechanism: Attention at L0 copies “hot” to the “dog” position. The MLP at L4 recognizes the compound at the “dog” position (“fried” first appears in the MLP output). Attention at L5 broadcasts this from “dog” to “was” (H19, attn=0.91). The contrastive projection identified each of these stages; activation patching at three positions and multiple layers confirms the information flow.

4.2 Other disambiguation cases

The same method traces disambiguation in verb-object pairs and noun ambiguity:

"He caught a cold and" vs "He caught a fish and" (verb meaning changes):

- L12: "contracted, virus, contagious" (cold pole)
- L28: "fever, coughing, cough, flu" vs "proudly, reel, bait, trout"

"The bank was steep and" vs "The bank was closed and" (noun disambiguated by adjective):

- L12: "climb, inclined, steep" (terrain pole)
- L27–28: "bank, banks, banking" reappear in the contrastive projection — the noun's representation is revisited after the modifier has disambiguated it. This is compatible with a two-pass pattern (resolve meaning from the modifier, then update the noun representation), though the contrastive projection reads content, not mechanism.

"He was fired up and" vs "He was fired and" (particle changes meaning):

- L28: "excited, ready, energetic" (fired up) vs "sued, blacklist, lawsuits" (fired)

4.3 Scalar implicature: pragmatic inference in the residual stream

Lexical disambiguation (§4.1–4.2) traces how the model resolves word meaning. A different kind of distinction involves quantifiers: "Some of the students passed" and "All of the students passed" produce different continuations. We trace how the residual stream diverges between these two quantifiers.

Design: Contrast "Some of the students passed the exam, so" against "All of the students passed the exam, so" and read the contrastive projection at the final token.

Trajectory (some pole $+$, all pole $-$) (Table 10):

Layer	Some pole reads	All pole reads
L8	might, may	except, together
L20	Others, another	everyone, except, Everyone
L28	others, Others, another	everyone, Everyone, everybody, except

Table 10: Contrastive trajectory for the some/all quantifier contrast (positive pole = some, negative pole = all).

The negative pole consistently reads "everyone, everybody, except" from L8 onward — tokens associated with "all" contexts. This reflects the divergent prediction trajectories of the two quantifiers: the "all" context primes totality tokens, and the contrastive projection surfaces this divergence.

Whether this reflects an internal "not everyone" computation or simply the output distribution difference between the two prompts cannot be determined from the contrastive readout alone.

Cross-content consistency. The same some/all contrast across 6 noun phrases (students, cookies, houses, employees, books, countries) produces pairwise cosine 0.12–0.45 (mean 0.30). The negative pole consistently reads totality words ("everyone, all, except, none") across all noun phrases, while the positive pole varies with content. Using multi-contrast triangulation across noun phrases, the shared component reads "Others, Other, another" vs "all, everyone, none, except, everybody" — both poles token-shaped.

Scalar gradient. Projecting five quantifiers (None, Few, Some, Most, All) onto the some↔all axis at L28 (Table 11):

Quantifier	Projection
None	+27
Few	+14
Some	+84
Most	+13
All	0 (baseline)

Table 11: Projection of five quantifiers onto the some↔all axis at L28. "Some is the outlier, consistent with its status as the pragmatically marked quantifier.

The gradient is not monotonic: "Some" is the outlier, projecting far beyond any other quantifier. "Few" and "Most" project similarly despite being semantically opposite. This is consistent with "some" being the pragmatically marked quantifier — the one that generates scalar implicature — rather than the scale reflecting quantity.

Explicit vs bare. Contrasting "Some of the students passed" against "Some but not all of the students passed" produces a non-trivial Δh (norm 57–59 at L28). The $-$side reads "some, Some, not, neither, but, BUT" — the explicit restriction changes the representation. The bare "some" computes partial implicature (the "everyone/except" signal), but adding "but not all" adds further restriction beyond what the bare form computes.

5 Replicating landmark findings

5.1 Indirect object identification

The IOI circuit (Wang et al., 2023) identifies how GPT-2 resolves which name a pronoun refers to. We replicate the core finding with the contrastive method.

Design: Contrast "John and Mary went to the store. John gave a book to" vs the same with names swapped. The IO name should appear in the contrastive projection at the prediction site.

Prompts (example):

- A: "John and Mary went to the store. John gave a book to" → Mary

- B: “Mary and John went to the store. Mary gave a book to” $\rightarrow$ John

Contrastive trajectory:

- L8: gender emerges (“himself, his” vs “herself, her”)
- L24: IO names appear (“Mary, Mary” vs “John, John”)
- L28: fully crystallized (“Mary, Mary, mary” vs “John, John, john”)

Per-head decomposition at L24: Decomposing the attention output by head identifies which heads carry the IO name. Three heads dominate consistently across three name pairs (John/Mary, Alice/Bob, Dan/Eve; Table 12):

Head	John/Mary (norm)	Alice/Bob	Dan/Eve	Content
H14	8.3	—	11.4	IO name (MD/Mary, de/draw)
H1	7.5	5.9	7.0	IO name (Mary, Bob, ever/ves)
H16	3.4	4.5	12.8	IO name (Mary, Bob, Eve)

Table 12: Per-head contrastive norm at L24 for the IOI name-swap contrast across three name pairs. Three heads carry the indirect-object name.

These heads have the largest contrastive norms at L24 and read IO-name tokens when projected through W_U. No other head exceeds norm 3.0 consistently. This identifies heads that carry the IO name signal — the same observational signature as the name-mover heads that Wang et al. (2023) found in GPT-2, here located in Phi-2 using only contrastive projection.

Why the name-swap contrast cannot test routing. Prompts A and B run the identical IOI algorithm — detect the duplicated subject, suppress it, copy the other name — and differ only in which tokens fill the roles. The shared algorithm cancels under subtraction, so the heads the contrast surfaces are name-content carriers, not the routing mechanism. Three tests sharpen what they are and whether they are causal.

Attention patterns confirm two genuine name-movers. At L24, H14 and H16 attend from the final position to the IO token (“Mary”), with 0.86 and 0.56 of their attention mass on it — reading the indirect object directly, the name-mover signature. H1, also flagged by contrastive norm, instead attends to the sequence-initial sink token (0.87): it is a false positive of the norm criterion that the attention pattern filters out. Pairing contrastive norm with attention direction separates the readers (H14, H16) from sink heads.

Ablation and denoising agree: the heads are not the causal carrier. Ablating all three heads at L24 (zeroing their pre-projection output at the last position) reduces P(Mary) from 0.696 to 0.685 — a 1.1% drop; individual ablation gives H14 -0.2%, H16 -0.9%, H1 none. Three control heads (H0, H5, H10) give a comparable 0.3% drop and the generation is unchanged. Denoising is the more sensitive complement: we corrupt the prompt by swapping the giver (P(Mary) = 0.011) and patch clean head outputs back at the final position, measuring how much of the clean–corrupt gap (toward 0.696) is restored. Patching the name-movers restores $\leq 1\%$; patching all 32 heads at the final position restores $\leq 2\%$ — at every layer from L20 to L28. Injection gives the same verdict for sufficiency: the name-movers' contrastive output (A$-$B) shifts P(Mary) in B from 0.026 to only 0.037 at $1\times$ and 0.067 at $3\times$, and patching all 32 heads A$\rightarrow$B reaches only 0.041. By contrast, injecting the full residual-stream Δh at L28 drives P(Mary) to 0.642. The decision is carried by the cumulative residual stream, not by any layer's attention write at the final position.

A structure-preserving contrast surfaces a name-invariant signal. Contrasting the IOI prompt against a no-duplicate baseline (giver = a third name, "Sam") keeps the duplicate-detection computation in the subtraction. This surfaces different heads — at L26 one reads the giver name ("Sam"), others read structural tokens — and the resulting delta does not project to clean tokens through W_U.

W_U-illegibility alone is weak evidence for structure, however: the logit lens is a single linear readout, and token content can be superposed in a basis it cannot decode. We test the claim directly with prompt design rather than a trained probe. Build a family of eight name-sets that share the IOI structure but vary the token identities (John/Mary, Alice/Bob, ...), and form both contrasts at END for each set. As depth increases the two diverge sharply: the name-swap (token) direction rotates with the names and collapses toward orthogonality across the family (mean pairwise cosine $+0.73$ at L20 $\rightarrow$ $+0.21$ at L24 $\rightarrow$ $+0.01$ at L28), whereas the no-duplicate direction stays aligned ($+0.83 \rightarrow +0.66 \rightarrow +0.37$). Averaging each contrast over the family — which cancels token-specific components and keeps the shared one — preserves 0.83 of the no-duplicate norm at L24 against 0.56 for the name-swap. The averaged name-swap survivor reads as W_U junk (its per-set reads are clean but different names — "Mary", "Sam" — that cancel), while the averaged no-duplicate survivor reads as identity-free recipient-role content ("charity, share, shop, others"). Position-resolved attention agrees: heads H12 and H20 read END$\rightarrow$duplicate-position across all eight name-sets (0.16–0.37), keying on the *position* regardless of which name fills it — a structural signature that never touches W_U.

The signal is therefore present, contrast-isolable, and structural rather than token-shaped — name-invariant where the name-identity content is not. (The surviving direction reads as the recipient role, which is broader than "inhibition" specifically; separating routing from recipient-role content would need a further control.)

Tracing the path by position-resolved patching. The corrupted (swapped) prompt differs from the clean one in a *single* input token — the second subject at position 9 (S2) — so all causal divergence originates there. Patching the full residual stream at one position from clean into the corrupted run, across layers, traces where the information lives (recovery of P(Mary), corrupt 0.011 $\rightarrow$ clean 0.696):

The IO position never matters — its token is identical in both prompts. The S2 position carries all of the recovery at L0 and loses it monotonically, while END gains it in lockstep; S2 and END jointly recover $\sim$100% at every layer. The causal signal transfers from S2 to END between L16 and L28, with the crossover (END-only overtaking S2-only) at L18–20.

Attention identifies the heads that perform the transfer. At L18–20, heads attending from END to

Patched position	L0	L8	L12	L16	L20	L24	L28
IO (pos 3)	0%	0%	0%	1%	0%	0%	0%
S2 (pos 9)	100%	90%	79%	59%	16%	6%	0%
END	0%	0%	0%	1%	37%	59%	98%
S2 + END	101%	95%	88%	87%	96%	97%	100%

Table 13: Position-resolved patching of the IOI prompt (recovery of P(Mary), corrupt → clean). The causal signal transfers from S2 to END between L16 and L28.

S2 dominate (H10, H12, H20, H31) — they move the duplicate-subject signal forward. At L22–26, heads attending to the IO token take over (H3, H14, H16, H0) — the name-movers that read and write the indirect object. This is the S-inhibition → name-mover structure of the IOI circuit, spread across roughly eight heads and eight layers rather than concentrated in a few.

This also resolves the apparent null of the per-head tests. The END residual after L22 recovers 59–98% of the prediction, but no single head's END output does (≤2%): the signal arrives via the cumulative S2→END transfer, carried in the residual stream, so ablating or patching any one head leaves the others to reconstruct it.

Summary. Contrastive projection, with attention confirmation, correctly identifies the genuine name-mover heads (H14, H16) and their read site (the IO token), and exposes H1 as a sink-head artifact of the norm criterion. Position-resolved patching then traces the full path: an identifiable origin (S2), a transfer window (L18–22), and two head populations (S2-readers then name-movers). The IOI computation in Phi-2 is distributed but fully traceable — no single head is necessary (ablation) or sufficient (per-head patching), because the decision rides the cumulative residual stream. This differs from GPT-2, where Wang et al. (2023) found the circuit concentrated in a few heads; here the same algorithm is spread across many heads and layers, and the contrastive method plus position-resolved patching recovers its structure end to end.

Accuracy: 30/30 across 5 name pairs × 3 templates on Phi-2 (100%), 30/30 on Pythia-1.4B (100%), 29/30 on Pythia-410M (97%).

5.2 Factual recall

Design: Contrast prompts requiring different factual answers.

"The Eiffel Tower is located in" vs "The Colosseum is located in":

- L20: "French, France" vs "ancient, Roman, Alexandria"
- L28: "French, Paris" vs "Rome, Roma, Gladiator"

Per-head decomposition: At L24, H13 carries the largest contrastive norm (3.9, reads "France, French, Paris"). At L28, the signal distributes across multiple heads (H21, H7, H0), consistent with the factual content spreading from a concentrated source to a broader representation.

Causal path trace. To test whether this concentration is causal rather than correlational, we use a single-token contrast — "The capital of **France** is" ($\rightarrow$ Paris) vs "The capital of **Japan** is" ($\rightarrow$ Tokyo) — which differ in exactly one input token, the subject. All causal divergence therefore originates at the subject position, so position-resolved patching (restore the clean residual stream at a chosen position and layer into the corrupt run; measure recovery of P(Paris)) traces where the answer is built and when it moves. CLEAN P(Paris) = 0.806, CORRUPT = 0.001.

layer	patch SUBJ only	patch END only	SUBJ+END
L0–L16	$\sim$100%	$\sim$0%	$\sim$100%
L18	99%	10%	100%
L20	74%	42%	100%
L22	50%	57%	100%
L24	8%	89%	100%
L28	0%	99%	100%

Table 14: Position-resolved patching for factual recall (recovery of P(Paris)). The answer is held at the subject token through the early layers, then handed to END around L22.

The answer lives entirely at the subject position through L16–L18 (a flat plateau: patching the subject alone fully restores Paris), then drains to the END position over L18–L28, with the handoff crossover at **L22**. Patching subject and END jointly recovers $\sim$100% at every layer — the signal rides the cumulative residual along a single subject$\rightarrow$END path, with no third site. The plateau-then-move shape matches the subject-MLP enrichment Meng et al. (2022) and Geva et al. (2023) report for factual recall, and the handoff sits a few layers later than IOI's (L22 vs L18–20).

Position-resolved attention from END names the carriers: attention to the subject peaks at L24, led by **H13** (0.31) and H16 (0.45) — and H13 is the same head the contrastive-norm decomposition independently flagged as reading "France/French/Paris." The norm criterion ("this head's output differs") and the causal trace ("this head moves the subject signal to END") converge on the same unit.

Generality. The subject$\rightarrow$END structure replicates across six entity pairs (France/Paris, Japan/Tokyo, Italy/Rome, Spain/Madrid, Egypt/Cairo, Russia/Moscow), every one with clean P $\geq$ 0.81, corrupt P $\approx$ 0, and a crossover at L22–L24.

Different factual domains: "The capital of France is" vs "The capital of Japan is" — the contrastive reads "Paris, France, French" on the France pole and "Japanese, Japan, Tokyo" on the Japan pole. Each entity's associated knowledge cluster appears on its respective side.

5.3 Factual recall vs hallucination

When the model hallucinates — producing a confident but fabricated answer for a fictional entity — the contrastive projection reveals what the model has retrieved: specific facts for real entities, and nothing beyond name fragments for fictional ones.

Design: We construct matched pairs where one prompt elicits genuine recall and the other elicits hallucination, keeping the frame identical (Table 15):

Real prompt	Prediction	Fictional prompt	Prediction
Nikola Tesla, born in 1856, invented the	Tesla coil…	Ludvig von Vogelkirche, born in 1859, invented the	first practical electric motor…
Marie Curie, born in 1867, discovered	the elements polonium and radium…	Helena Brandström, born in 1871, discovered	a new species of moth…

Table 15: Matched real/fictional entity prompts; both elicit confident, specific predictions.

Both sides produce confident, specific answers. But the contrastive projection at L28 reads different content on each pole (Table 16):

Pair	Real pole (L28)	Fictional pole (L28)
Tesla vs Vogelkirche	Tesla, alternating, Altern, electric	Vog, v, von, Von
Curie vs Brandström	radio, Radio, Radiation, radioactive	Brand, M, H, a

Table 16: Contrastive projection at L28: the real pole reads factual associations, the fictional pole only name fragments.

The real pole reads *factual associations* (Tesla $\rightarrow$ alternating current, Curie $\rightarrow$ radioactivity). The fictional pole reads *name fragments* (Vog, von, Brand) — the model has retrieved nothing about the entity beyond the tokens of its name and their cultural associations.

Confirmation via entity-vs-generic contrast. Contrasting each entity against a bare frame ("A person, born in 1856, invented the") isolates what the entity name adds. Tesla minus generic reads "Tesla, alternating, AC" at L28 — specific factual content. Vogelkirche minus generic reads "Vog, von, Von" — only name tokens. The hallucinated entity's representation contains no factual content beyond the name itself.

Contrastive norm. The relative norm $\|\Delta h\| / \|h\|$ at L28 is systematically larger for real-vs-fictional pairs (mean 0.98) than real-vs-real pairs (mean 0.70). Two real entities both occupy a shared subspace of factual-entity representations, constraining their difference; a fictional entity lacks factual content and drifts outside this subspace, producing a larger contrastive norm against any real entity.

Entropy. Real entities predict with lower entropy (H = 1.5–2.5) than fictional ones (H = 3.4–6.3), consistent with the model having specific knowledge to draw on. But entropy alone cannot distinguish "confident recall" from "confident hallucination" — the mountain case illustrates this: Mount Silverhorn (fictional) predicts "2,856 meters" with H = 2.8, similar to Mount Everest's H = 1.5. The contrastive projection shows the difference: Everest's pole reads "Nepal, Tibet, Himal" (geographic knowledge), while Silverhorn's reads "1300, 1100, 1200" (number-range priors calibrated to "Southern Alps," not specific factual recall).

The contrastive projection does not find a "hallucination flag." It reads what the model has retrieved, and for hallucinated entities, the retrieval is empty of factual content — only name tokens and contextual priors remain.

5.4 Successor heads and temporal structure

All successors are predicted correctly, including wrap-arounds: "After Saturday comes" → Sunday; "After Sunday comes" → Monday; "After December comes" → January.

Contrastive trajectory: Contrasting consecutive successors (e.g., "After Monday comes" vs "After Tuesday comes"), the contrastive projection at L24 reads the input day name on its respective pole ("Monday" tokens positive, "Tuesday" negative). By L28 the successor day appears: "Tuesday" on the Monday pole, "Wednesday" on the Tuesday pole. The successor computation is visible as the transition from input-day to output-day content across layers.

Per-head decomposition at L28: Decomposing the contrastive signal by attention head identifies H11 as the dominant successor head. Its contrastive norm spikes at the Saturday→Sunday and Sunday→Monday transitions (Table 18):

Day pair	H11 norm	Next-largest head
Mon→Tue	2	H20 (3)
Tue→Wed	3	H20 (3)
Wed→Thu	3	H20 (2)
Thu→Fri	4	H20 (5)
Fri→Sat	4	H25 (3)
Sat→Sun	**11**	H12 (4)
Sun→Mon	**11**	H12 (5)

Table 17: Contrastive trajectory contrasting a counterfactual capital assertion against the veridical one. From L20 the dominant signal is correction ("Actually).

H11's norm triples at these two transitions; we do not have evidence for what distinguishes them from the weekday transitions. The same head dominates month-pair contrasts at L28, with norms 3–9 across all twelve transitions.

H11 dominates the successor signal but does not carry it alone — H20 and H25 contribute at comparable norms for some day pairs (e.g., H20 norm=5 at Thu→Fri vs H11 norm=4).

A separate PCA analysis of the raw hidden states (not using the contrastive projection) finds circular geometry for months and days; this is reported in the supplementary materials as it does not use the paper's method.

5.5 Parametric recall resists counterfactual assertion

When in-context information contradicts parametric knowledge, does the model override its stored facts? We test this by asserting a counterfactual capital and then querying: "The capital of France is Rome. The capital of France is."

The model does not override. Across four country/capital pairs (France/Rome, Japan/London, Germany/Madrid, Italy/Vienna), the model predicts the correct parametric answer with high confidence despite the counterfactual assertion (P(Paris) = 0.77 vs 0.81 baseline; P(Tokyo) = 0.87 vs 0.30 baseline). Even with stronger framing ("It is well established that the capital of France is Rome"), the model predicts Paris at 0.46. Only fictional framing ("In the wizarding world") partially weakens parametric recall (Paris 0.31, Rome 0.18).

The contrastive trajectory reads correction, not acceptance. Contrasting the counterfactual assertion against the veridical one (Table 17):

Layer	Counterfactual pole reads	Parametric pole reads
L8	similarly, similar	[subword fragments]
L20	Actually, actually, actual	[subword fragments]
L24	actually, Actually, correct	centrally, other
L28	Actually, Greece, Paris	[comparison tokens]

Table 18: Per-head contrastive norm at L28 for consecutive-day successor contrasts. H11 dominates and spikes at the two week-boundary transitions.

From L20 onward, the dominant signal is "Actually, actually, correct" — the model prepares a correction rather than accepting the override. At L28, the Paris token logit is *higher* on the counterfactual side (+9.6) than on the veridical side — the model strengthens its parametric recall when presented with a contradicting assertion. The same holds for all four country pairs.

Distance weakens correction. Inserting distractor text between the assertion and query partially weakens the correction reflex: with a medium distractor (∼20 words), P(Paris) drops to 0.38 and P(Rome) rises to 0.16. The contrastive trajectory at L28 shifts from reading "Actually" to reading the asserted city ("Rome, Italy, Madrid"), suggesting the correction signal decays with distance while the asserted content persists.

Connection to hallucination (§5.3). The correction response — "Actually, it is Paris" — requires the model to have retrieved the parametric fact. The hallucination cases in §5.3 show that fictional entities produce no factual associations in the contrastive projection. The correction circuit and the factual-retrieval circuit may share structure: both require that the model has specific knowledge to draw on, and both are visible in the contrastive trajectory as the presence or absence of factual content at mid-to-late layers.

6 Contrastive axis taxonomy

Using 2×2 factorial designs (crossing two binary axes, e.g., past/future × happy/sad), we measure whether each axis produces a consistent contrastive direction across content. Consistency is the cosine between the axis direction extracted from two different content fillers. We test 15 axes across four models (Pythia-410M, Pythia-1.4B, Phi-2, Phi-4).

6.1 Three tiers of axis consistency

Axes partition into three tiers:

Tier 1: Cross-family, token-readable (cos > 0.7 in all four models). Eight axes exceed 0.7 consistency in all four models, though some weaken at scale: code/natural language (0.88–0.95), positive/negated (0.84–0.90), past/future (0.81–0.89), English/French (0.83–0.98, declines at scale), assignment/equality-test (0.79–0.92, improves with scale), CAPS/lowercase (0.71–0.87), doubt/certainty (0.71–0.96, weakens at scale), claim/question (0.74–0.83). These axes are near-orthogonal to their content fillers (mean |cross-cosine| 0.06–0.14).

Tier 2: Partially consistent (cos 0.5–0.8 in some models). Four axes depend on model family or content: formal/informal (strong in Pythia, weak in Phi), thought/speech (moderate, reads as subjective-evaluation vs reporting), cause/effect (weakens at scale).

Tier 3: Not a direction (cos < 0.5 in all models). Four axes never form a consistent direction: animate/inanimate (0.34–0.48), active/passive (0.17–0.65, entangles with content), literal/metaphorical (0.19–0.35, worsens at scale), salient-entity/generic (collapses to -0.12 at Phi-4).

6.2 Negation is not cancellation

Projecting different negation types onto the contrastive `not` direction (Table 19):

Negation	Pythia-1.4B	Phi-2
not	+43	+69
never	+36	+63
no longer	+35	+54
not never	**+35**	**+66**
rarely	+22	+49

Table 19: Projection of different negation types onto the contrastive `not` direction. Double negation ("not never) does not cancel.

Double negation ("not never") does not cancel — it projects at 81–95% the strength of single "not." The model represents "not never" as emphatic negation. Each negation type has its own token readout: `not` reads as "not, NOT, Not"; `no longer` reads as "gone, now, replaced" (temporal displacement); `rarely` reads as "seldom, usually, often" (frequency scale).

6.3 Metaphor is processed by domain routing, not a flag

The literal/metaphorical axis has the lowest consistency (0.19–0.35) because metaphor is not a single direction. Instead, each metaphorical use routes to its target domain (Table 20):

Causal verification (leave-one-out). For each word, we extract the routing direction from 3 of 4 literal-metaphorical pairs (pairwise cos 0.64–0.88) and test injection on the held-out 4th pair. Note that the extraction and injection sets overlap — the same pairs are used for both — so this tests causal relevance of the direction, not held-out generalization. Injecting the cold literal direction

Contrast	Literal pole reads	Metaphor pole reads
cold: ice vs reception	higher, Celsius, Fahrenheit	tense, atmosphere, tension, mood
sharp: knife vs criticism	blade, blades, stainless	tone, sarcastic, condescending
bright: lamp vs student	blinding, overpowering, intensity	proud, gifted, amazed, grades
heavy: boulder vs news	exceed, load, exert	mood, tense, gloomy, somber

Table 20: Literal vs metaphorical contrasts: each metaphorical use routes to its own target domain rather than to a shared figurativity direction.

($+1.0\times$) into "The reception was extremely cold. The atmosphere was" shifts predictions from "chilly" (social) to "below, freezing" (temperature). Reverse injection ($-1.0\times$) into "The ice was extremely cold. The temperature was" shifts from "below" to "tense" (social). The same pattern holds for sharp, bright, and heavy: each direction bidirectionally flips between the word's literal and metaphorical domain.

Dose-response. The cold domain flip is graded: the metaphorical context shifts through chilly $\rightarrow$ freezing $\rightarrow$ below as injection magnitude increases from $0.25\times$ to $2.0\times$. Top-1 prediction flips at $0.65\times$ of natural magnitude.

Routing directions are per-domain-pair, not a universal axis. The cross-domain cosine matrix reveals the structure (Table 21):

	cold	sharp	bright	heavy
cold	1.00	+0.28	+0.06	+0.58
sharp	+0.28	1.00	+0.03	+0.37
bright	+0.06	+0.03	1.00	+0.14
heavy	+0.58	+0.37	+0.14	1.00

Table 21: Cross-domain cosine matrix of the four metaphor routing directions. Cold and heavy share structure; bright is nearly orthogonal to the rest.

Cold and heavy share substantial structure (cos 0.58) — both route between a physical-measurement domain and an emotional domain. Sharp and cold share less (0.28). Bright is nearly orthogonal to all others (0.03–0.14) because it routes to intelligence, a different target domain. The routing direction is determined by the pair of domains being mapped between, not by a shared literalness feature. Cross-domain injection confirms this: injecting the cold routing direction into a sharp-metaphorical context pushes predictions toward temperature tokens ("below, lower"), not toward blade tokens — each direction routes to its respective literal domain.

This explains why literal/metaphorical has Tier 3 consistency: there is no single metaphor axis because there is no single target domain. Probing classifiers trained on cold-metaphor would transfer to heavy-metaphor (same target domain) but not to bright-metaphor (different target domain).

6.4 Axis consistency predicts causal potency

The tier classification connects to the causal tests in §3.2. The positive/negated axis (Tier 1, cos 0.84–0.90) produces a direction that bidirectionally flips true/false predictions when injected. The eatability direction — extracted by the same multi-contrast averaging used for axes — flips food/animal predictions at 30% of natural magnitude.

We tested whether axis consistency predicts whether the W_U readout of an axis functions as a frame-forcing token. For six contrastive scenarios, we extracted the axis direction, read its top tokens through W_U, and tested those tokens as adjective modifiers on 15 nouns (Table 22):

Scenario	Axis consistency	W_U top token	Override rate
flying (vs parked)	0.74	"future", "enabled"	87–93%
stolen (vs displayed)	0.67	"rightfully"	93%
deadly (vs harmless)	0.57	"deadly"	100%
burning (vs standing)	0.59	"got", "lis"	0–7%
frozen (vs fresh)	0.44	"oop", "paradox"	7%

Table 22: Axis consistency vs frame-forcing override rate. Above ∼0.6 consistency, the W_U readout yields tokens that act as frame-forcers when used as modifiers.

When axis consistency exceeds ∼0.6, the W_U readout surfaces tokens that function as frame-forcers when used as modifiers. Below ∼0.5, the direction is still causal when injected (the burning direction shifts predictions from "built" to "evacuated") but its W_U readout does not produce usable tokens. The contrastive direction is causally relevant in both cases; only its token-readability depends on consistency.

7 Discussion

7.1 What the method does

Contrastive projection desuperposes the residual stream along one axis of variation. The subtraction cancels what is shared between two inputs; the W_U projection reads the remainder in token space. The method locates where a distinction first appears (per-position reading), which component writes it (attention vs MLP decomposition), and which head carries it (per-head decomposition).

7.2 What the readout means

The contrastive projection reads the model's internal representations in token space. W_U assigns token labels to directions in the residual stream; these labels name the features the model has detected and is operating on, verified by three independent tests: causal injection (§3.1), dose-response (§3.2), and MLP neuron correspondence (§3.3).

When the readout produces uninterpretable tokens, the target signal is typically still present but superposed with other signals. Multi-contrast triangulation (§2.4) recovers the causal content in every case we tested. This does not guarantee that all model computation is token-readable —

it establishes that for the 18 contrasts tested, the causal content is token-shaped once properly desuperposed.

7.3 Superposition and contrast design

The quality of the readout depends on the contrast. A well-chosen pair that varies one axis produces a clean readout. A poorly chosen pair — or one where the target concept is compositional and superposed with pair-specific content — produces tokens that reflect the superposition, not the target. Multi-contrast averaging addresses this for cases we tested, but we have not established how many baselines are sufficient in general, or whether all model computations can be desuperposed by this technique.

7.4 Token readability and causal relevance

Across 16 contrastive pairs, 4 layers, and 32 attention heads (1774 head-level measurements), heads whose contrastive output reads as real words have higher causal alignment on average (mean $|\cos|$ 0.044 for 5/5-readable vs 0.025 for 0/5-readable; Pearson $r = +0.15$). The most causally aligned individual head contributions in our sample are token-readable, and no unreadable head exceeds $|\cos| = 0.16$. The sample is limited and the correlation is modest.

7.5 Relationship to circuit analysis

The contrastive projection locates phenomena; circuit analysis explains them. Wang et al. (2023) mapped the IOI circuit in GPT-2 via path patching. Our method identifies heads with the same functional signature in Phi-2 (H14, H1, H16 at L24) by contrastive norm, but does not establish their causal roles. The method is closest to causal tracing (Meng et al., 2022): it identifies where information concentrates, then hands off to heavier tools for causal verification.

7.6 Limitations

- **Curated pairs, not sampled.** All demonstrations use hand-constructed minimal pairs. The multi-contrast triangulation uses hand-selected baselines.
- **LayerNorm bypassed.** We skip the final LayerNorm, so W_U receives vectors at the wrong scale. Token rankings are empirically invariant to this choice, but raw contrastive norms are not comparable across layers due to residual-stream norm growth.
- W_U **readability not guaranteed.** The difference of two states was never trained for W_U projection. Token labels at intermediate layers are W_U's nearest-neighbour assignments. The MLP neuron correspondence (§3.3) provides independent evidence that these labels match internal model structure for the cases tested.
- **Smoothness is not W_U-specific.** Trajectory coherence (consecutive-layer cosine) is a property of Δh, not of W_U.

- **Exploratory, not causal.** The per-position trace and per-head decomposition identify large contributors, not causes. We verify causality via activation patching for the hot dog case (§4.1) and via injection recovery for four cases (§3.1), but the per-head decomposition (§5) is observational.
- **Desuperposition coverage.** We tested multi-contrast triangulation on 6 cases and MLP neuron correspondence on 18 cases. We do not know whether all model computations can be desuperposed by contrastive subtraction, or how many baselines are sufficient in general.
- **Per-position reading requires tokenization alignment.** The read position must correspond to the same structural role in both inputs.
- **Model coverage.** Primary mechanistic results (§4) on Phi-2 only. IOI replicates on Pythia models. The axis taxonomy (§6) covers four models and shows both consistent and model-dependent patterns.

8 Related work

Contrastive activation methods. RepE (Zou et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024) use matched-pair subtraction for steering. Du et al. (2026) apply it to R1-style reasoning models. Li et al. (2024) use contrastive pairs to study internal representations of truthfulness. Ma et al. (2026) contrast contextualized and non-contextualized logits to locate and rectify conflict-inducing layers; they intervene to resolve knowledge conflicts, whereas we use the same contrast only to read. We use the same arithmetic for reading, with per-position tracing and attention/MLP decomposition.

Logit and tuned lens. nostalgebraist (2020), Belrose et al. (2023). Project individual states through W_U. Pal et al. (2023) (Future Lens) show a single hidden state also carries information about tokens several positions ahead. The contrastive projection reads the content that differs between two inputs — a different subspace from what the logit lens shows for either input individually.

Probing and linear representations. Linear probes (Belinkov, 2022) train classifiers on hidden states to detect features. Our method achieves a related decomposition without training — the contrastive subtraction is a zero-shot linear probe along the axis defined by the input pair. Burns et al. (2023) extract truth directions from contrast pairs using unsupervised methods; our multi-contrast triangulation is a related technique applied to arbitrary semantic axes rather than truth specifically.

Superposition and sparse autoencoders. Elhage et al. (2022) characterized superposition in toy models. Bricken et al. (2023) and Templeton et al. (2024) use sparse autoencoders to decompose superposed representations into monosemantic features. Our multi-contrast triangulation achieves a related decomposition — isolating one signal from superposition — using paired inputs rather than learned dictionaries. The MLP neuron correspondence (§3.3) connects to the neuron-level analysis in this literature but uses contrastive gating rather than unsupervised feature discovery. Lange et al. (2026) report that the sparsity objectives used to train cross-layer transcoders can incentivize circuits that match behavior while rewriting deep computation into shallow form. Our reading has no learned dictionary and no sparsity penalty, so it is not subject to that particular incentive; it also does not recover circuit topology, and we make no broader faithfulness claim.

Circuit analysis and causal tracing. Wang et al. (2023) reverse-engineered the IOI circuit in GPT-2 via path patching. Meng et al. (2022) used causal tracing to localize factual associations. Gould et al. (2024) identified successor heads via attention pattern analysis. Conmy et al. (2023) automated circuit discovery via ACDC. Our method recovers the key observational findings from these papers (which layers, which heads, which content) but does not establish causality — it is an exploratory complement to these causal techniques.

Factual recall and knowledge editing. Geva et al. (2023) traced factual associations to MLP layers. Hernandez et al. (2024) studied linearity of relational knowledge. Our hallucination detection (§5.3) and parametric correction (§5.5) results connect to this line: the contrastive method reads what was retrieved, and for fictional entities the retrieval is empty.

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